

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: **Computer Organization and Architecture** | Course Code: **21CS34** | Credits: **3**

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Demonstrate Booth's Multiplication Algorithm to multiply multiplicand $M = 7$ (0111) and multiplier $Q = -3$ (1101) showing register states (A, Q, Q-1) after each arithmetic cycle. [8 Marks]

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Q2. Explain Cache Memory mapping techniques: (i) Direct Mapping, (ii) Associative Mapping, and (iii) Set-Associative Mapping with address format breakdown and hit/miss evaluation. [8 Marks]